

Netflix Content Analysis using Python Pandas

Problem Statement:

Traditional spreadsheet tools often struggle with large datasets that require complex string manipulation and conditional filtering. With over 8,000 entries and multiple missing data points, the Netflix titles dataset requires a programmatic approach to ensure data quality. The goal of this project was to demonstrate the power of Python Pandas in automating the 'Data Wrangling' process.

Loading Data And Looking At First Five Rows

```
[3]
✓ Os
import pandas as pd
df = pd.read_csv('netflix_titles.csv')
print(df.head())
```

	show_id	type	title	director	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

		cast	country	
0		NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...		South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...		NaN	
3		NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...		India	

	date_added	release_year	rating	duration	
0	September 25, 2021	2020	PG-13	90 min	
1	September 24, 2021	2021	TV-MA	2 Seasons	
2	September 24, 2021	2021	TV-MA	1 Season	
3	September 24, 2021	2021	TV-MA	1 Season	
4	September 24, 2021	2021	TV-MA	2 Seasons	

	listed_in	
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
2	Crime TV Shows, International TV Shows, TV Act...	
3	Docuseries, Reality TV	
4	International TV Shows, Romantic TV Shows, TV ...	

	description
0	As her father nears the end of his life, filmm...
1	After crossing paths at a party, a Cape Town t...
2	To protect his family from a powerful drug lor...
3	Feuds, flirtations and toilet talk go down amo...
4	In a city of coaching centers known to train I...

Checking For Missing Values And Data Types

[4]

✓ 0s



```
print(df.info())
```

▼

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
None
```

Description:

In this initial phase, I imported the pandas library and loaded the Netflix dataset. I used the `.head()` function to inspect the first five rows of the data. This allowed me to understand the structure of the dataset, including columns such as `show_id`, `type`, `title` and `duration`. I noticed several missing values (NaN) in the 'director' and 'cast' columns which signaled that data cleaning would be required."

Data Cleaning

[10]

✓ 0s

```
df['country'] = df['country'].fillna('Unknown')
df.dropna(subset=['date_added', 'rating'], inplace=True)
df['date_added'] = pd.to_datetime(df['date_added'])
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month_name()
```

Description:

Real-world data is rarely perfect. In this section, I performed data cleaning. I identified missing values across the dataset and handled them by filling empty country slots with 'Unknown' and dropping rows with critical missing information in the 'date_added' and 'duration' columns. This ensures that my subsequent analysis and mathematical calculations are accurate and free from errors.

Separating Movies And TV Shows

[11]

✓ 0s



```
movies_df = df[df['type'] == 'Movie'].copy()
shows_df = df[df['type'] == 'TV Show'].copy()
```

Description:

To conduct a more focused analysis, I used Boolean filtering to split the main dataset into two distinct Data Frames. One for Movies and one for TV Shows. This separation is crucial because these two content types use different metrics for length.

Dropping Rows Where Duration Is Missing And Manipulating The Data

```
[13]
✓ Os
movies_df = movies_df.dropna(subset=['duration'])

movies_df['duration_min'] = movies_df['duration'].str.replace(' min', '').astype(int)

long_movies = movies_df[movies_df['duration_min'] > 120]
print(f"Number of movies longer than 2 hours: {len(long_movies)}")
```

Number of movies longer than 2 hours: 1142

Description:

In this step, I performed data transformation on the 'duration' column. Since the values were stored as strings, I used string manipulation to strip the text and converted the column to an integer type. This transformation allowed me to perform numerical analysis such as filtering for movies longer than two hours.

Finding The Most Common Number of Directors

```
[14]
✓ Os
top_directors = df['director'].value_counts().head(10)
print(top_directors)
```

...

director	
Rajiv Chilaka	19
Raúl Campos, Jan Suter	18
Marcus Raboy	16
Suhas Kadav	16
Jay Karas	14
Cathy Garcia-Molina	13
Martin Scorsese	12
Jay Chapman	12
Youssef Chahine	12
Steven Spielberg	11

Name: count, dtype: int64

Description:

With a clean and transformed dataset, I was able to extract meaningful insights. I calculated the number of movies longer than 120 minutes and identified the top 10 directors with the most content on the platform. This demonstrates how Pandas can turn thousands of rows of raw data into summarized and actionable information.

Creating Table Where The Most Action Movies Are After 2015

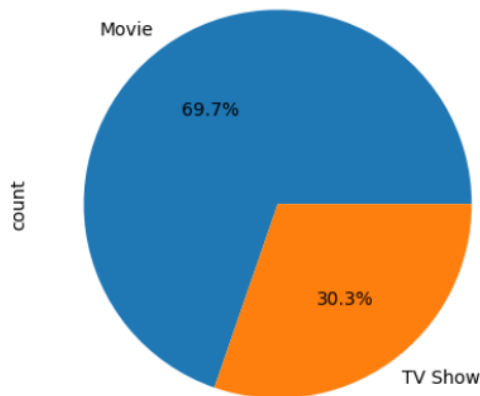
```
[17]
✓ Os
action_content = df[(df['listed_in'].str.contains('Action', case=False)) & (df['release_year'] > 2015)]
country_counts = action_content.groupby('country').size().sort_values(ascending=False).head(5)
print("Top 5 Countries for Action:")
print(country_counts)
```

```
... Top 5 Countries for Action:
country
United States    156
India            44
Unknown         42
China           21
Japan           21
dtype: int64
```

Simple Pie Chart

```
[19]
✓ Os
df['type'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```

```
<Axes: ylabel='count'>
```



Description:

I also make the table where 'Action' Movies are released after 2015. To conclude the project, I created a visual representation of the content distribution. By using the .plot() method, I visualized the ratio of Movies to TV Shows which providing a clear high-level summary of Netflix content library.